SOFTWARE REQUIREMENTS SPECIFICATION (SRS)
Order Dispatch & Delivery Tracking System (ODDTS)

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Table of contents

S.No	Content	Page No.
0.	Preface (Document Information)	3
1.	Introduction	3
2.	Overall Description	4
3.	System Objectives	5
4.	UML Models and Diagrams	6
5.	User Requirements	21
6.	System Requirements	22
7.	Functional Requirements	22
8.	Non-Functional Requirements	22
9.	Domain Constraints and Business Rules	23
10.	Out of Scope	23
11.	Risks and Challenges	23
12.	Compliance and Ethics	24
13.	Limitations of Current Version	24
14.	Future Enhancements	24
15.	Appendix	24

0. Preface

0.1 Intended Audience

This document is prepared for system users, developers, faculty evaluators, and project reviewers who are involved in understanding, building, and assessing the Order Dispatch & Delivery Tracking System (ODDTS).

0.2 Purpose of Document

This Software Requirements Specification (SRS) describes the functional and non-functional requirements of the Order Dispatch & Delivery Tracking System (ODDTS). It clearly explains what the system is expected to do and the constraints under which it must operate.

1. Introduction

1.1 Purpose

This document specifies the software requirements for the Order Dispatch & Delivery Tracking System (ODDTS). It describes the system's functionality, constraints, and quality attributes.

1.2 Scope

ODDTS is a web-based full-stack application designed to manage the complete lifecycle of delivery orders from order creation to final delivery confirmation. The system supports multiple user roles and ensures secure, persistent, and rule-based logistics operations.

The system will evolve across three semesters in functionality and technical maturity.

1.3 References

Sommerville, I., Software Engineering – Requirements Engineering
Course Material: Full Stack Development classroom slides

2. Overall Description

2.1 Product Perspective

ODDTS is a standalone full-stack web application consisting of a frontend user interface, backend application server, and a persistent database (MySQL or MongoDB).

2.2 Product Functions (High Level)

The system provides the following major functions:

- Order creation and management
- Dispatch and assignment of delivery agents
- Delivery tracking and confirmation
- Notification and exception handling
- Reporting and analytics
- Authentication and authorization

2.3 Stakeholders

The stakeholders of the system include:

- Merchants
- Dispatch Managers
- Delivery Agents
- Customers
- System Administrators
- Development Team
- Faculty Evaluators

2.4 User Classes and Characteristics

Merchant – Creates and manages delivery orders
Dispatch Manager – Assigns delivery agents and monitors deliveries
Delivery Agent – Executes deliveries and updates order status
Customer – Tracks orders and confirms delivery
Administrator – Manages system configuration and business rules

2.5 Operating Environment

- Web browsers such as Chrome, Firefox, and Edge
- Backend server using Node.js and Express

- Database using MySQL
- Hosting on cloud or institutional server

2.6 Design and Implementation Constraints

- The system shall be implemented as a web-based application
- Persistent database storage is mandatory
- Role-based access control must be enforced
- RESTful API architecture shall be used

2.7 Assumptions and Dependencies

- Users have access to internet and modern browsers
- Initial deployment targets small logistics organizations
- GPS-based tracking may be added in future versions

3. System Objectives

3.1 Overall Objective

To design and develop a scalable, secure, and reliable Order Dispatch & Delivery Tracking System that supports end-to-end logistics operations from order creation to delivery confirmation, using modern full-stack technologies and persistent data storage.

3.2 Specific Objectives

- Model Real-World Logistics Workflows:
  To analyze and model real-world dispatch and delivery processes by identifying key actors, responsibilities, and workflows using UML diagrams and structured requirements.
- Provide Core Order and Dispatch Management Functions:
  To implement essential functionalities such as order creation, agent assignment, delivery status updates, and exception handling in a structured and rule-based manner.
- Ensure Secure and Role-Based System Access:
  To enforce authentication and role-based authorization so that each actor (Merchant, Dispatch Manager, Delivery Agent, Customer, Administrator) can access only permitted system functions.
- Support Tracking, Notifications, and Customer Interaction:
  To provide shipment tracking, delivery notifications, and feedback/complaint mechanisms for improved transparency and user experience.
- Maintain Data Integrity and System Reliability:
  To ensure persistent data storage, audit logging, and validation mechanisms that guarantee consistency, traceability, and reliability of system operations.

- Design for Scalability and Future Enhancements:
  To develop the system architecture in a modular and extensible manner so that advanced features such as analytics, mobile applications, and intelligent routing can be integrated in future versions.

4. UML Models and Diagrams

This section presents the UML diagrams used to model the functional behavior and workflows of the Order Dispatch & Delivery Tracking System (ODDTS). These diagrams help visualize actors, system interactions, and internal processes based on real-world logistics operations.

The diagrams ensure clarity in system design and act as a reference for developers and evaluators to understand the flow of operations.

4.1 Use Case Diagram

The Use Case Diagram illustrates the interactions between the system and its actors. It identifies the major functionalities provided by the system and maps them to the respective actors.

The main actors include:

- Merchant
- Dispatch Manager (Hub Manager)
- Delivery Agent
- Customer
- Administrator

The Use Case Diagram covers core functionalities such as:

- Order Management
- Dispatch & Assignment Management
- Delivery Execution
- Exception & Return Handling
- Shipment Tracking & Notification
- Feedback & Complaint Management
- Reporting & Monitoring
- System Administration & Security

This diagram provides a high-level functional overview of the system.

USE CASE



These diagrams help in understanding the control flow, validations, and exception scenarios involved in real-world delivery processes.

4.2 Activity Diagrams

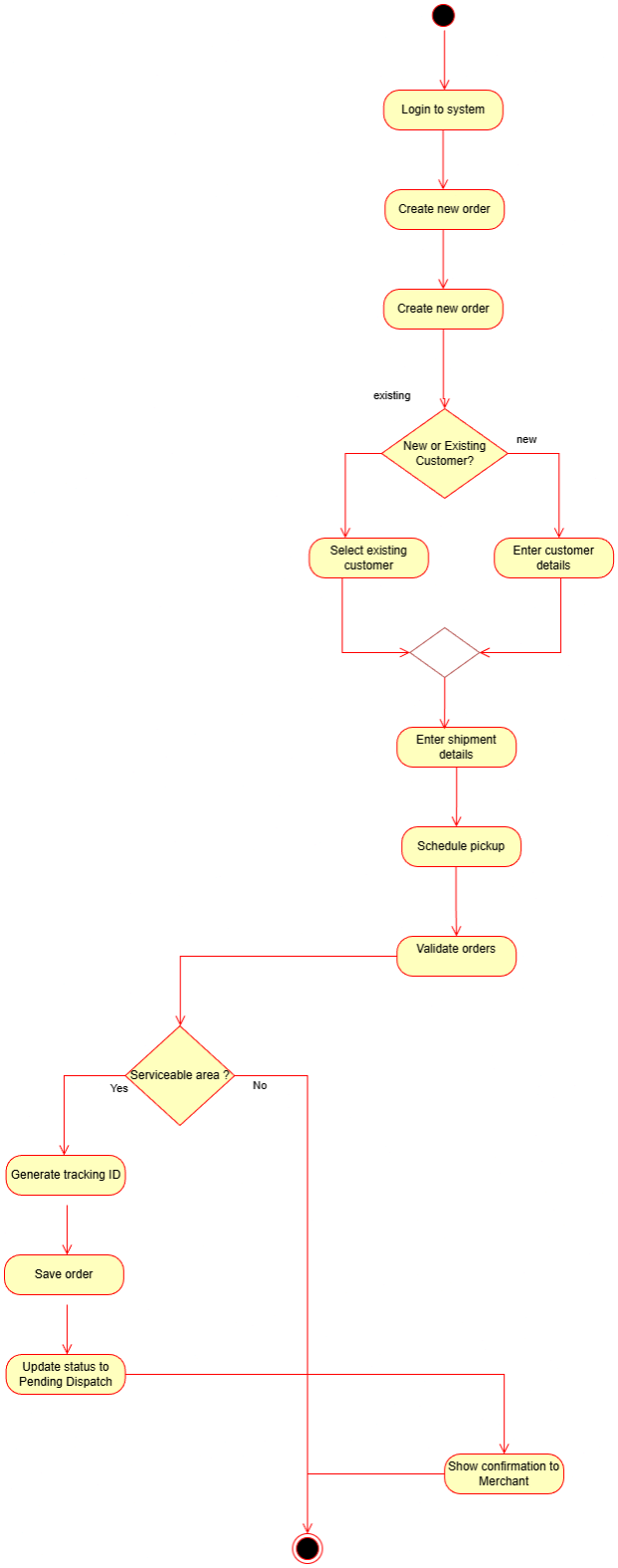
The Activity Diagrams describe the detailed workflows of major system processes. These diagrams represent step-by-step execution of tasks and decision points involved in logistics operations.

Key workflows modeled using Activity Diagrams include:

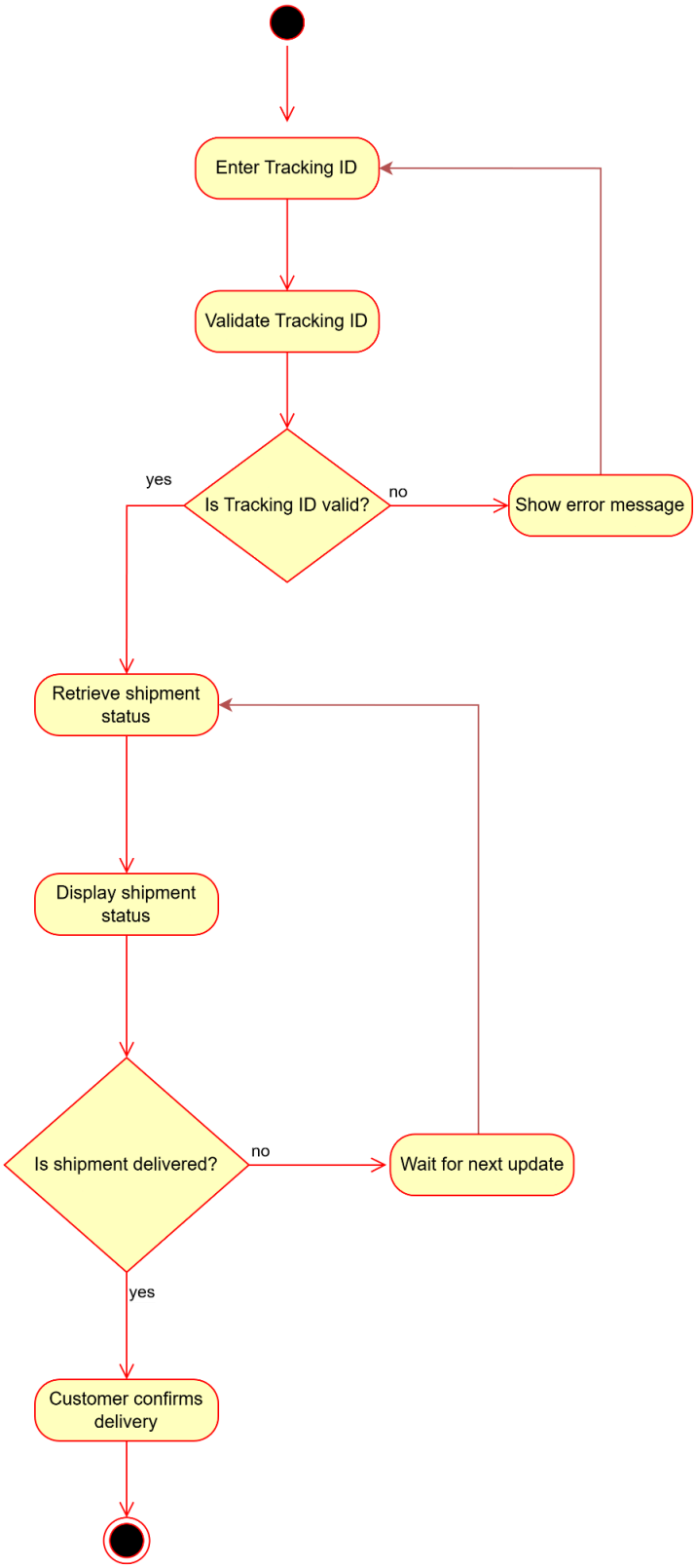
1. Order Management
2. Shipment Tracking
3. Delivery Execution
4. Exception & Return Handling
5. Shipment Tracking & Notification
6. Feedback & Complaint Management
7. Reporting & Monitoring
8. System Administration & Security

These diagrams help in understanding the control flow, validations, and exception scenarios involved in real-world delivery processes.

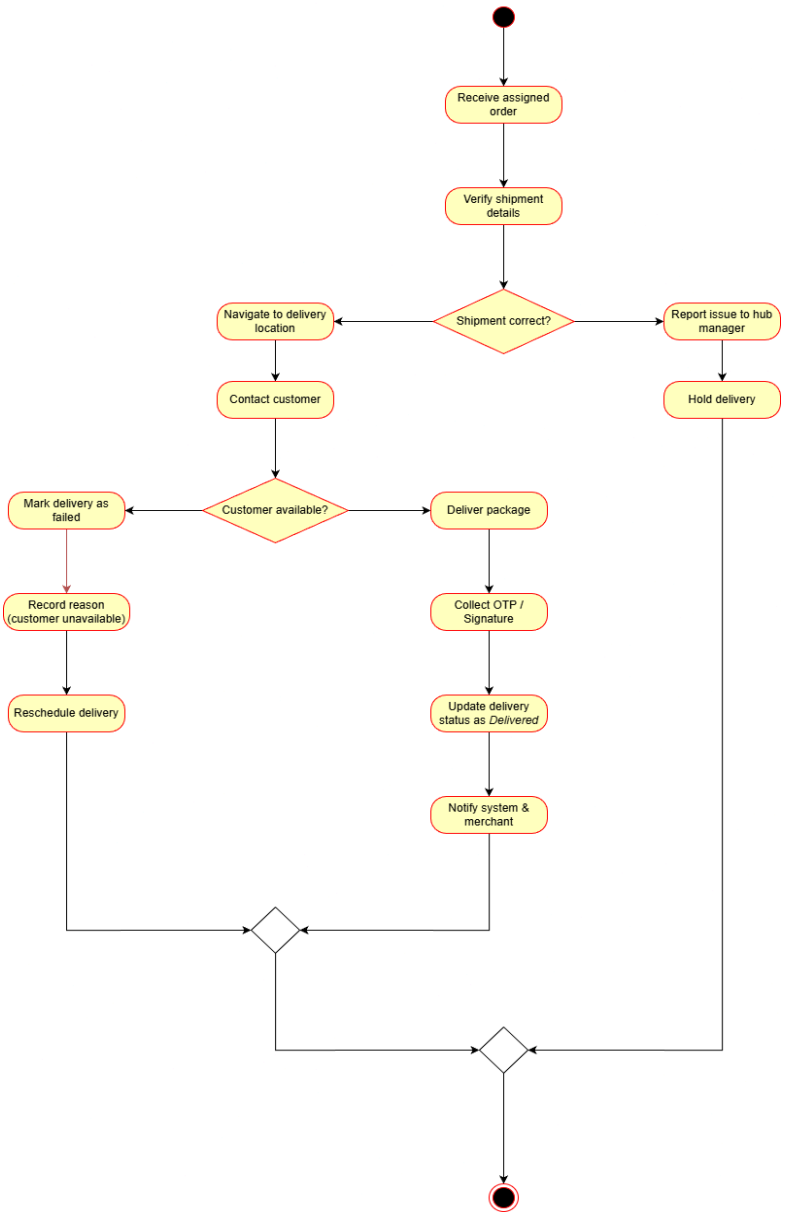
4.2.1 Order Management



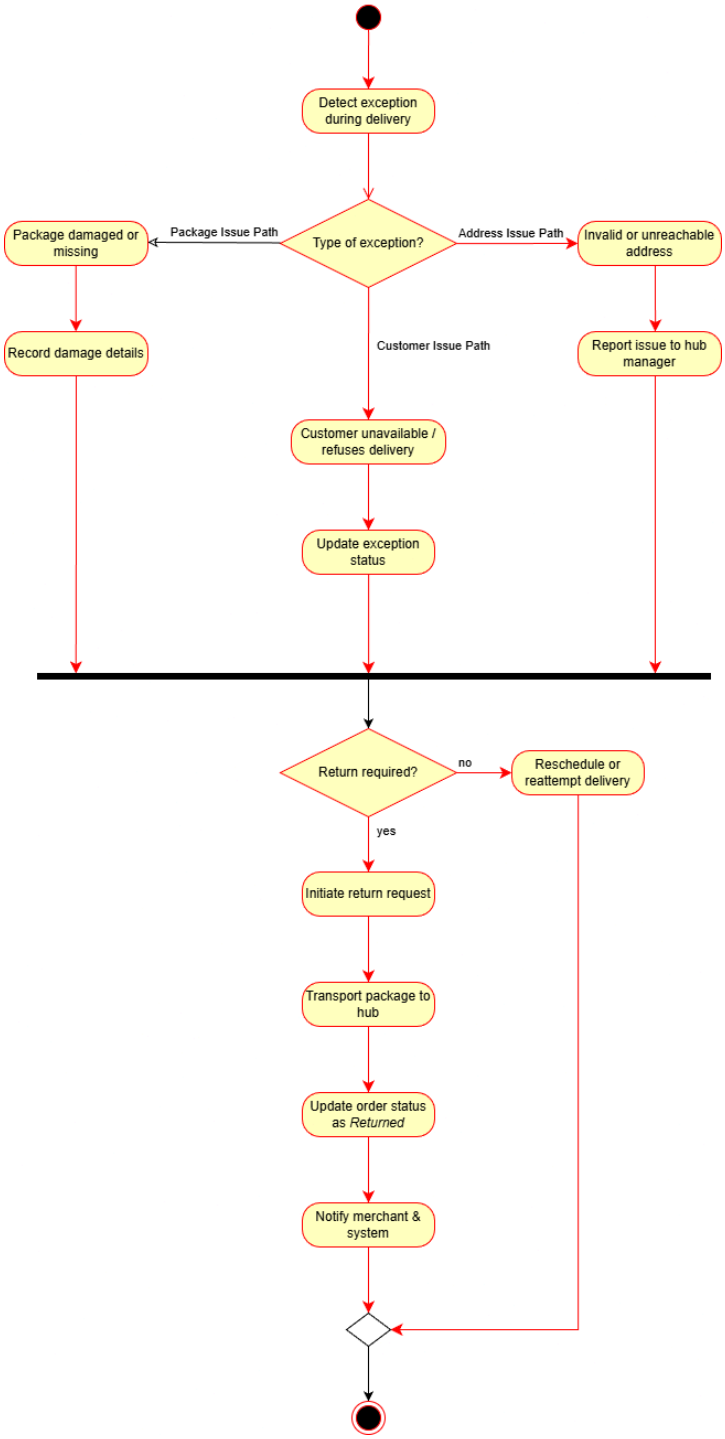
4.2.2 Shipment Tracking



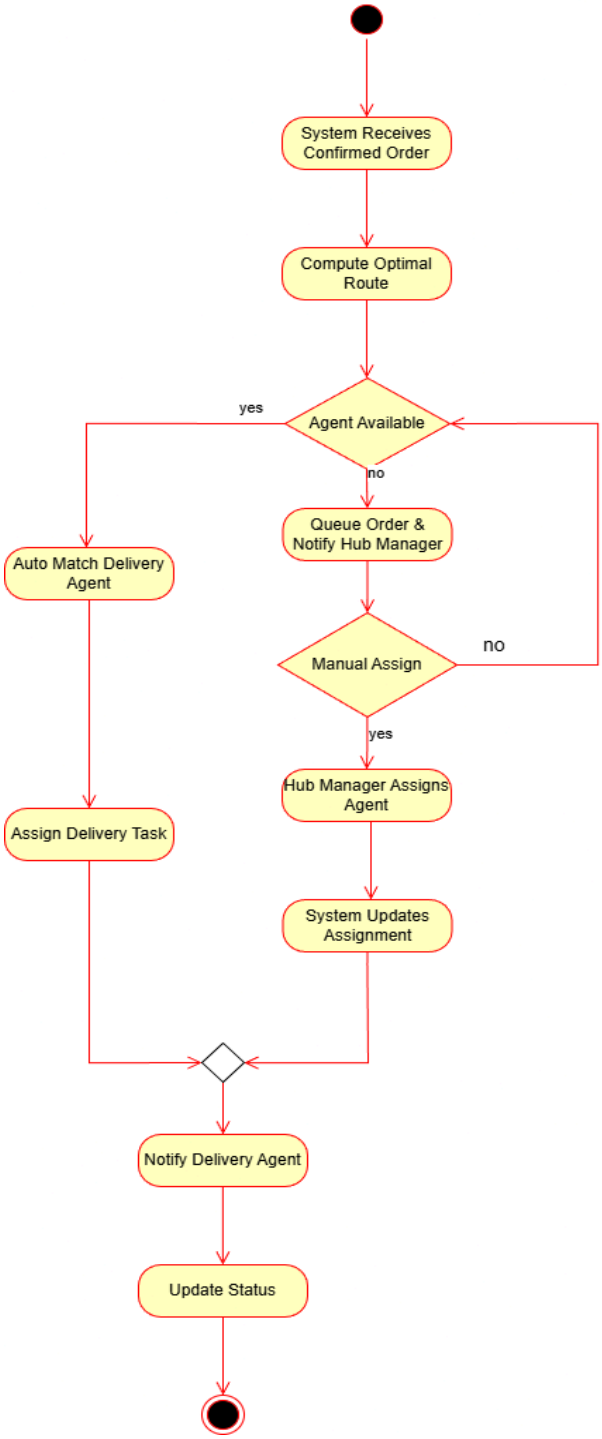
4.2.3 Execute Delivery



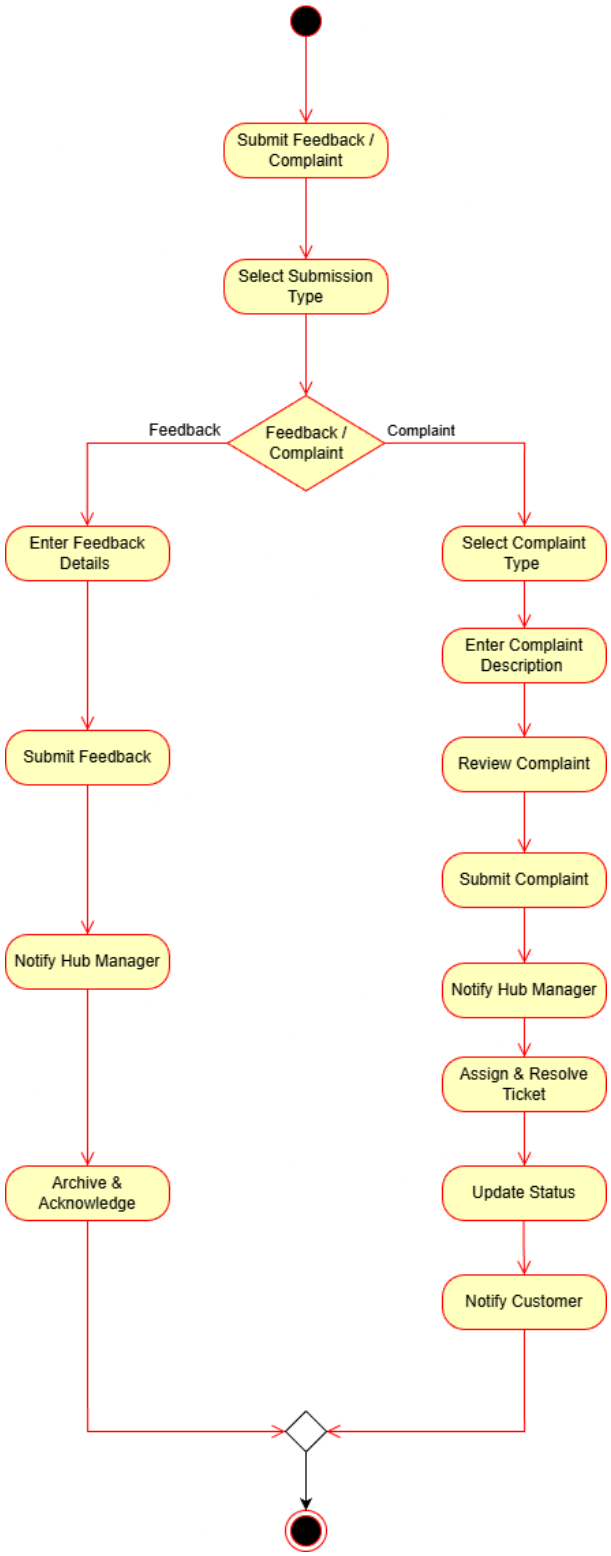
4.2.4 Exception & Return Handling



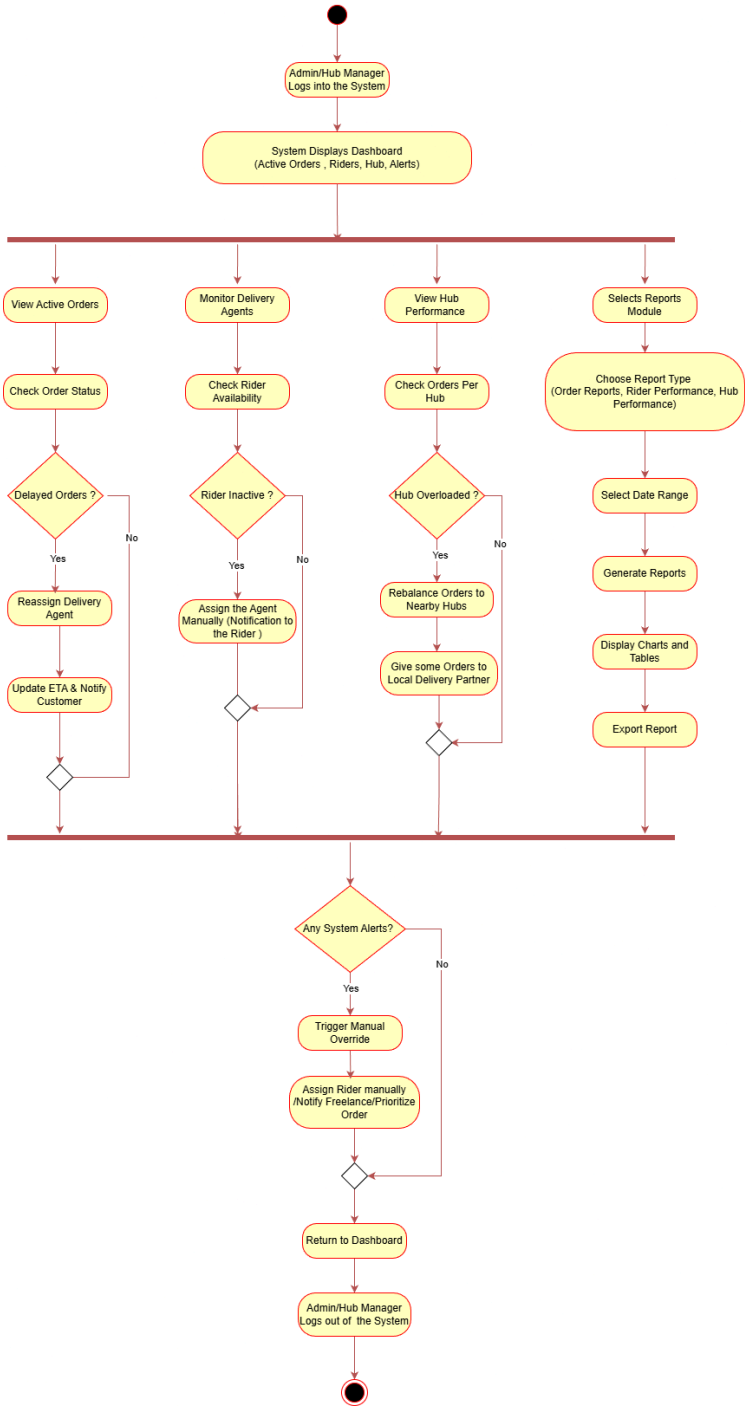
4.2.5 Dispatch & Assignment Management



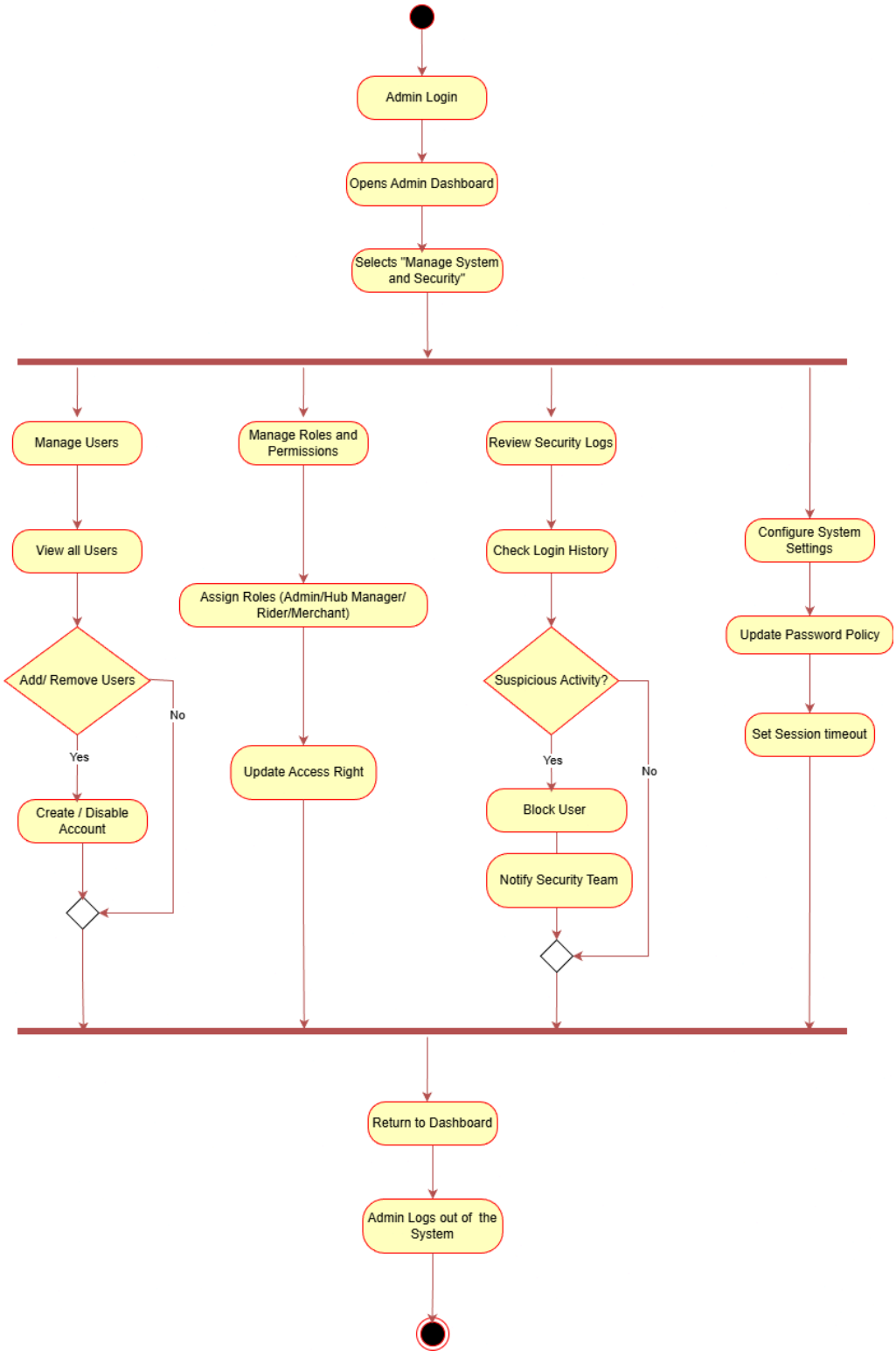
4.2.6 Feedback & Complaint Management



4.2.7 View Reports and Monitor Operations



4.2.8 System Administration & Security



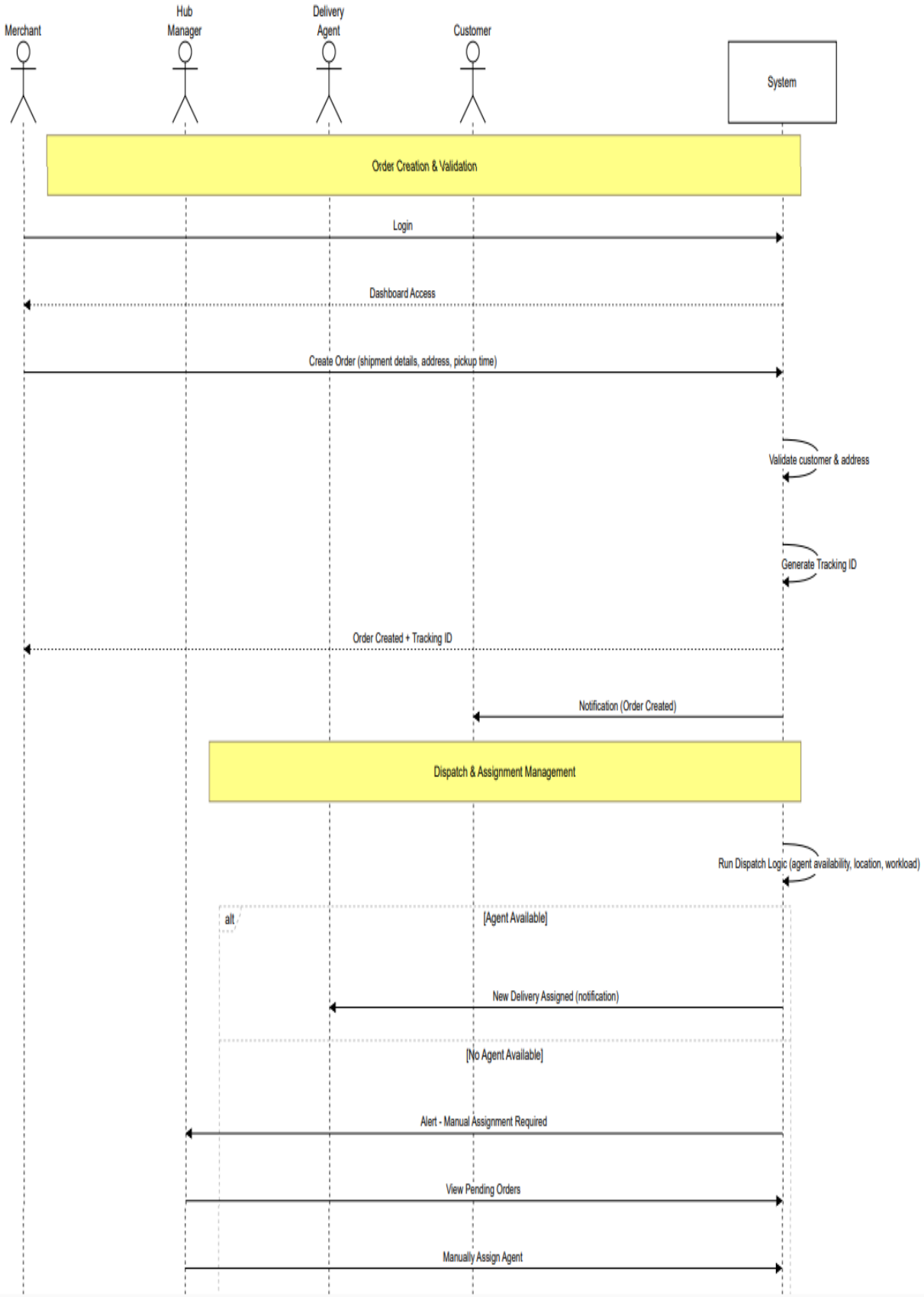
4.3 Sequence Diagram

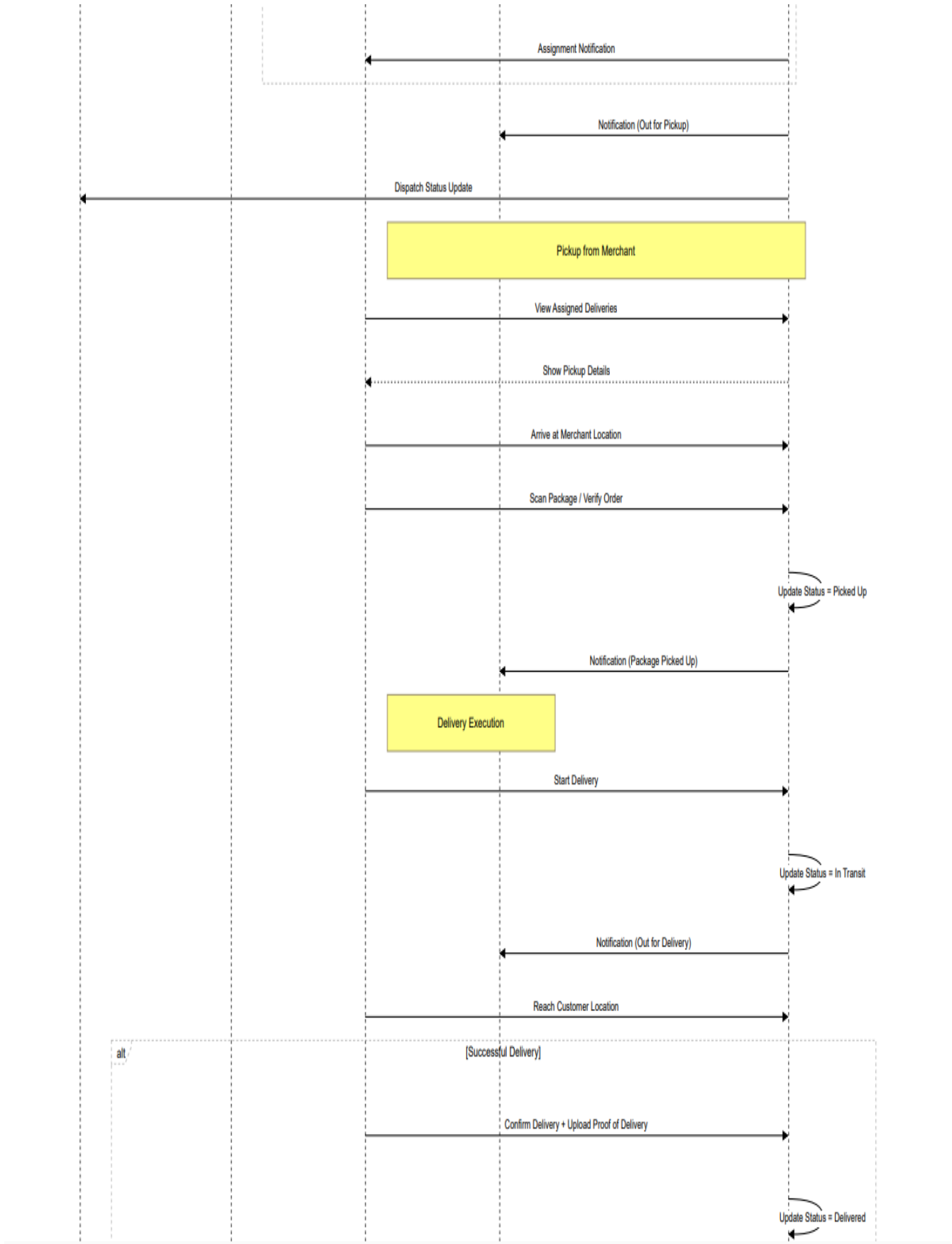
The Sequence Diagram represents the time-ordered interaction between actors and system components during the execution of key use cases.

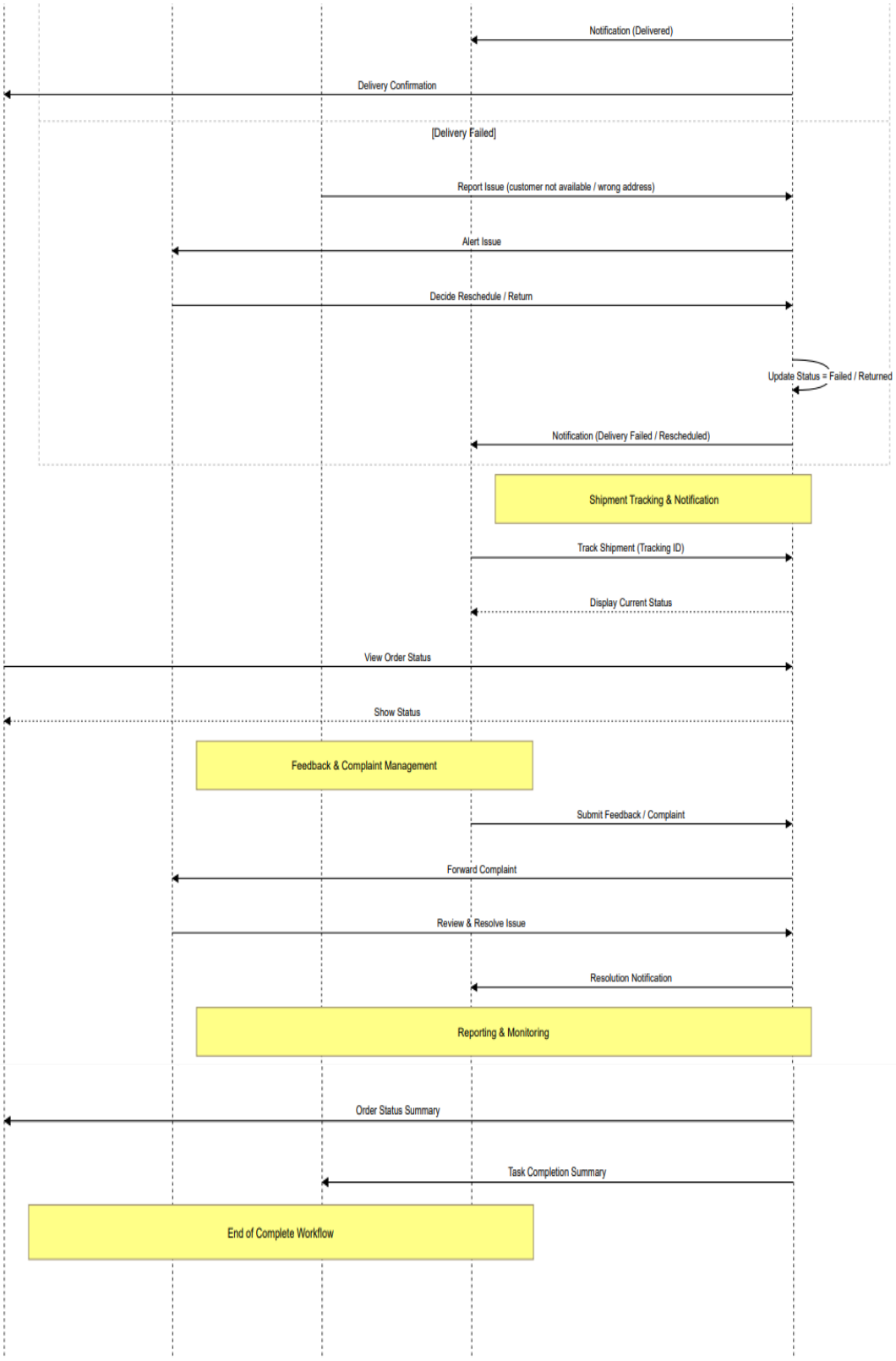
It demonstrates:

- How a merchant creates an order
- How the system assigns a delivery agent
- How the delivery agent updates delivery status
- How the customer tracks shipments and receives notifications
- How exceptions and feedback are handled

This diagram ensures that system interactions are logically ordered and aligned with the functional requirements.


Merchant
Hub Manager
Delivery Agent
Customer
System
Order Creation & Validation
Login
Dashboard Access
Create Order (shipment details, address, pickup time)
Validate customer & address
Generate Tracking ID
Order Created + Tracking ID
Notification (Order Created)
Dispatch & Assignment Management
Run Dispatch Logic (agent availability, location, workload)
alt
[Agent Available]
New Delivery Assigned (notification)
[No Agent Available]
Alert - Manual Assignment Required
View Pending Orders
Manually Assign Agent


Assignment Notification
Notification (Out for Pickup)
Dispatch Status Update
Pickup from Merchant
View Assigned Deliveries
Show Pickup Details
Arrive at Merchant Location
Scan Package / Verify Order
Update Status = Picked Up
Notification (Package Picked Up)
Delivery Execution
Start Delivery
Update Status = In Transit
Notification (Out for Delivery)
Reach Customer Location
alt
[Successful Delivery]
Confirm Delivery + Upload Proof of Delivery
Update Status = Delivered


Notification (Delivered)
Delivery Confirmation
[Delivery Failed]
Report Issue (customer not available / wrong address)
Alert Issue
Decide Reschedule / Return
Update Status = Failed / Returned
Notification (Delivery Failed / Rescheduled)
Shipment Tracking & Notification
Track Shipment (Tracking ID)
Display Current Status
View Order Status
Show Status
Feedback & Complaint Management
Submit Feedback / Complaint
Forward Complaint
Review & Resolve Issue
Resolution Notification
Reporting & Monitoring
Order Status Summary
Task Completion Summary
End of Complete Workflow

5. User Requirements

5.1 Merchant Requirements

- UR-1 The system shall allow merchants to create delivery orders.
- UR-2 The system shall allow merchants to edit or cancel orders before dispatch.
- UR-3 The system shall allow merchants to schedule pickup times.
- UR-4 The system shall allow merchants to view order history.
- UR-5 The system shall generate confirmation for created orders.

5.2 Dispatch Manager Requirements

- UR-6 The system shall allow dispatch managers to view all pending orders.
- UR-7 The system shall allow dispatch managers to assign delivery agents.
- UR-8 The system shall allow dispatch managers to reschedule deliveries.
- UR-9 The system shall display delivery status dashboards.
- UR-10 The system shall generate performance and analytics reports.

5.3 Delivery Agent Requirements

- UR-11 The system shall allow agents to view assigned orders.
- UR-12 The system shall allow agents to update delivery status.
- UR-13 The system shall allow agents to report delivery issues using predefined reasons.
- UR-14 The system shall record delivery completion time.
- UR-15 The system shall allow agents to upload proof of delivery.

5.4 Customer Requirements

- UR-16 The system shall allow customers to track orders using order ID.
- UR-17 The system shall display current order status.
- UR-18 The system shall allow customers to confirm receipt.
- UR-19 The system shall allow customers to submit feedback.
- UR-20 The system shall allow customers to raise complaints.

5.5 Administrative Requirements

- UR-21 The system shall maintain an audit log of all actions.
- UR-22 The system shall notify users on order status changes.
- UR-23 The system shall allow configuration of business rules.

6. System Requirements

- SR-1 The system shall expose RESTful APIs for frontend-backend communication.
- SR-2 The system shall persist data using MySQL or MongoDB.
- SR-3 The system shall support concurrent users.
- SR-4 The system shall validate all user inputs.
- SR-5 The system shall log critical system events.

7. Functional Requirements

- FR-1 The system shall support user registration and login.
- FR-2 The system shall enforce role-based access control.
- FR-3 The system shall support order CRUD operations.
- FR-4 The system shall support delivery agent assignment.
- FR-5 The system shall support status updates (Pending, In Transit, Delivered, Failed).
- FR-6 The system shall provide order tracking to customers.
- FR-7 The system shall generate analytics and reports.
- FR-8 The system shall handle failed delivery workflows.
- FR-9 The system shall log all system activities.
- FR-10 The system shall support secure file upload for proof of delivery.

8. Non-Functional Requirements

8.1 Performance

- NFR-1 The system shall support at least 100 concurrent users.
- NFR-2 Average response time shall be less than 3 seconds.

8.2 Usability

- NFR-3 The system shall provide an intuitive and responsive user interface.
- NFR-4 The system shall provide meaningful error messages.

8.3 Reliability

- NFR-5 The system shall maintain 99 percent availability during business hours.
- NFR-6 The system shall log errors for diagnosis.

8.4 Security

- NFR-7 Passwords shall be securely hashed.
- NFR-8 JWT authentication shall be implemented.
- NFR-9 Role-based authorization shall be enforced.
- NFR-10 The system shall prevent SQL injection and XSS attacks.
- NFR-11 File uploads shall be validated and securely stored.

8.5 Scalability

- NFR-12 The system shall support database migration for growth.
- NFR-13 The system architecture shall allow horizontal scaling.

9. Domain Constraints and Business Rules

- An order cannot be marked Delivered unless marked In Transit.
- Only assigned delivery agents can update delivery status.
- Cancelled orders cannot be reassigned.
- All actions must be timestamped.
- Proof of delivery must be linked to completed orders.

10. Out of Scope

- Payment processing
- Inventory management
- Route optimization
- Third-party courier integration
- Real-time GPS tracking

11. Risks and Challenges

- Network failures
- Data entry errors
- Security threats
- User adoption issues
- Scalability limitations

12. Compliance and Ethics

- User data privacy must be protected.
- Access must be controlled based on roles.
- Audit logs must be maintained.
- Data protection principles must be followed.

13. Limitations of Current Version

- Designed for small-scale logistics operations.
- Does not support real-time GPS tracking initially.
- Limited analytics in early phases.

14. Future Enhancements

- GPS-based tracking
- AI-based route optimization
- Mobile application
- Third-party system integrations

15. Appendix (Glossary)

- Merchant – Business client who creates delivery orders
- Dispatch Manager – Logistics staff responsible for assigning and monitoring deliveries
- Delivery Agent – Individual responsible for delivering orders
- Customer – Recipient of the delivery order
- Administrator – User responsible for system configuration
- ODDTS – Order Dispatch & Delivery Tracking System
- JWT – JSON Web Token
- CRUD – Create, Read, Update, Delete